Reservoir computing over compressible signals outline

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Let $C_{1,0}([0,1];\mathbb{R}^d)$ be the space of absolutely continuous d-dimensional paths $X_{(\cdot)} : [0,1] \to \mathbb{R}^d$ with uniform topology. In this context, the reservoir computing phenomenon is usually understood as the phenomenon in which a sufficiently regular input/output system $F : C_{1,0}([0,1];\mathbb{R}^d) \to C_{1,0}([0,1];\mathbb{R}^m)$, given by $F(X)_t = \tilde{F}(X_{[0,t]})$ for some $\tilde{F} : C_{1,0}([0,1];\mathbb{R}^d) \to \mathbb{R}^m$ (causality), can be well approximated by the composition of: 1) a high dimensional linear state space system

$$dZ_t^X = \sum_{i=1}^{d} A_i' Z_t^X dX_t^i + B' dX_t, \text{started at } Z_0^X = C' X_0$$

$$Z_t \in \mathbb{R}^N, A_i' \in \mathbb{R}^{N\times N}, B' \in \mathbb{R}^{N\times d}, C' \in \mathbb{R}^{N\times d}$$

whose dynamics are governed by random projections A_i', B', C', and 2) an optimal choice of a linear readout $V \in \mathbb{R}^{m\times N}$. That is, for a given $\epsilon > 0$ the following holds:

$$\sup_{t\in[0,1]} ||F(X)_t - V Z_t^X||_2 < \epsilon, \forall X_{(\cdot)} \in K \subset C_{1,0}([0,1];\mathbb{R}^d)$$

with K satisfying e.g. compactness. Here and after, it suffices to consider the case $m = 1$. Moreover, for simplicity, we consider the case where $B = 0$, i.e. we have a linear (instead of affine) system.

There are many insightful results that shed light on why exactly this is possible. The following outline is an example of how we can arrive at this conclusion:

1 In continuous-time

1. (Characterization of solutions) First, it can be shown by the method of Picard iteration that Z_t^X with dynamics A_i, C has expansion

$$Z_t^X = \sum_{i=1}^{d} \sum_{\substack{n\in\mathbb{N}\\ 1\leq i_1\leq d,\ldots,1\leq i_n\leq d}} A_{i_1}\cdots A_{i_n} C_i X_0^i S(X)_{[0,t]}(i_1,\ldots,i_n) \tag{1}$$

in terms of its signature terms

$$S(\cdot)_{[0,t]}^n : X \mapsto (\int_{0<t_{i_1}<\ldots<t_{i_n}<t} dX_{t_{i_1}}\ldots dX_{t_{i_n}})_{1\leq i_1\leq d,\ldots,1\leq i_n\leq d} \in (\mathbb{R}^d)^{\otimes n}$$

at all times t.

2. (Closure of signature span) Second, the functions over paths of interest $\tilde{F} \in C(K) := \{\tilde{F} : K \to \mathbb{R} | \tilde{F}$ continuous w.r.t uniform convergence$\}$, are shown to be in the closure of the span of signature terms. This is done by means of a classical theorem of universal approximation using a Stone-Weierstrass argument. In view of this, the space $C_{1,0}([0,1];\mathbb{R})$ can be shown to be point-separable via the signature transform. More precisely, the terms $\{S(X)_{[0,1]}^n\}$ can be regrouped (in increasing order) to form an element of the tensor algebra $T((\mathbb{R}^d)) = \prod_{n=0}^{\infty}(\mathbb{R}^d)^{\otimes n} \supset \bigoplus_{n=0}^{\infty}(\mathbb{R}^d)^{\otimes n} =: T(\mathbb{R}^d)$, with $T(\mathbb{R}^d)$ admitting a dual space $T(\mathbb{R}^d)^* = \prod_{n=0}^{\infty}((\mathbb{R}^d)^{\otimes n})^* \cong \prod_{n=0}^{\infty}((\mathbb{R}^d)^*)^{\otimes n} \cong T((\mathbb{R}^d)^*)$, the latter having an action defined by applying the functional in $((\mathbb{R}^d)^{\otimes n})^*$ to the corresponding tensor in $(\mathbb{R}^d)^{\otimes n}$ and summing across $n = 0, 1, 2, \ldots$. Separability comes in the form that for any two $X, Y \in C_{1,0}([0,1];\mathbb{R}^d)$, there exist $f \in T(\mathbb{R}^d)^*$ of the form $g \circ \pi_{\leq N}$ where $g \in T(\mathbb{R}^d)^*$ and $\pi_{\leq N} : \bigoplus_{n=0}^{\infty}(\mathbb{R}^d)^{\otimes n} \to \bigoplus_{n=0}^{N}(\mathbb{R}^d)^{\otimes n}$ is the truncation operator, such that

$$f((S(X)_{[0,1]}^0, S(X)_{[0,1]}^1, \ldots)) \neq f((S(Y))_{[0,1]}^0, S(Y)_{[0,1]}^1, \ldots)).$$

Along with a few other arguments, this allows us to apply the Stone-Weierstrass theorem to conclude that $\Phi(K) := \{\phi : X \in K \mapsto g \circ \pi_{\leq N}(S(X)_{[0,1]}) | N > 0, g \in T(\mathbb{R}^d)^*\} \subset C(K)$ is dense in $C(K)$. In particular, for an input-output system F of the kind above, we can obtain

$$\forall t, \forall X \in K, F(X)_t = F(\tilde{X}_{[0,t]}) \overset{\epsilon-\text{close}}{=} g \circ \pi_{\leq N}(S(X_{[0,t]})_{[0,1]}) = g \circ \pi_{\leq N}(S(X)_{[0,t]})$$

for some $g, N > 0$.

3. (Random state matrices) We are not done yet, since we have yet to

exhibit some solution Z_t^X which, after a linear readout, will yield a linear combination of signature terms close to $F(X)_t$. In view of this, note that the expansion in 1 gives us access to the signature terms of $X_{[0,t]}$, on the conditon that we can come up with a "selection operator." For instance, to retrieve $X_0^i S(X)_{[0,t]}(i_1, ..., i_n)$ it suffices to find a vector $v \in \mathbb{R}^N$ such that $\langle v, A_{i_1} \cdots A_{i_n} C_i \rangle = \mathbf{1}_{\{(i_1,...,i_n)=(j_1,...,j_m)\}}$. In fact, Cirone et al (Cirone, 2024) showed that for some choice of random matrix ensembles $\mathbb{U}^{N\times N}, \mathbb{V}^{N\times d}$, sampling $A'_i \in \mathbb{U}^{N\times N}, C' \in \mathbb{V}^{N\times d}$, one has the asymptotic bound

$$||\frac{1}{N}\langle A'_{i_1}, ..., A'_{i_n} C'_i, A'_{j_1}, ..., A'_{j_m} C'_j \rangle - \mathbf{1}_{\{(i,i_1,...,i_n)=(j,j_1,...,j_m)\}}||_{L^2(\mathbb{P})} = \mathcal{O}(1/\sqrt{N})^2 2^{\frac{|I|+|J|}{2}}(|I|+|J|)!! \quad (2)$$

It remains to 1) apply the argument that $\phi(K) \subset C(K)$ is dense to find $g \in T(\mathbb{R}^d)^*$ such that $g \circ \pi_{\leq K}(\bar{S}(\cdot)_{[0,t]})$ is within an ϵ-error of $F(\cdot)_t$ in $|\cdot|$ over K for any input path X and time t. Afterward, 2) remarking that for some integer μ, $\mathbb{R}^\mu \cong \pi_{\leq K} T(\mathbb{R}^d)$, we can represent the action of $g \circ \pi_{\leq K}$ on $\pi_{\leq K} S(X)_{[0,t]}$ as a dot product between some coefficient vector $(\alpha_I)_{|I|\leq K} \in \mathbb{R}^\mu$ and $S(X)_{[0,t]})(I)_{|I|\leq K} \in \mathbb{R}^\mu$, where I denotes a multi-index of indices in $\{1, ..., d\}$. Then, 3) craft a vector $v \in \mathbb{R}^N = (1/N)\sum_{i=1}^d \sum_{|I|\leq K} \gamma_I^i \prod_{j\in I} A'_j C'_i$ which, when applied to Z_t^X with dynamics A'_i, C' hopefully yields an expression

$$\sum_{i=1}^d \sum_{|I|\leq K} \langle v, \prod_{j\in I} A'_j C'_i \rangle X_0^i S(X)_{[0,t]}(I)$$

that is close enough to

$$\sum_{i=1}^d \sum_{|I|\leq K} \gamma_I^i X_0^i S(X)_{[0,t]}(I)$$

which in turn can be molded into

$$\sum_{|I|\leq K} \alpha_I S(X)_{[0,t]}(I)$$

2 In discrete-time

The discrete case, as treated by Cuchiero et al., can be resolved with a mildly different outline. The "path space" considered is then $\ell^\infty(\mathbb{Z}^{\leq 0}; \mathbb{R}^d) =$

$\{\mathbf{z} \in (\mathbb{R}^d)^{\mathbb{Z}^{\leq 0}} : ||\mathbf{z}||_\infty := \sup_{t\leq 0} ||\mathbf{z}_t||_2 < \infty\}$, with "uniform topology" replaced by the norm topology induced by the "fading memory" norm $||\mathbf{z}||_w := \sup_{t\leq 0} ||\mathbf{z}_t||_2 w_{-t}$, where $(w_k)_{k\geq 0}$ is any strictly decreasing, positive sequence with $w_0 = 1$. In what follows, K_M is the closed ball of radius M w.r.t. the base norm $||\mathbf{z}||_\infty$.

1. (Closure of signature span) A causal, time invariant $U : K_M \subset \ell^\infty(\mathbb{Z}^{\leq 0}; \mathbb{R}^d) \to K_L \subset \ell^\infty(\mathbb{Z}^{\leq 0}; \mathbb{R})$ satisfying $U(\mathbf{0}) = 0$ that is analytic as a map between open sets can be expanded into a Taylor series, which at time $t \leq 0$ is polynomial in the variables $\mathbf{z}_t, \mathbf{z}_{t-1}, ...$. Knowledge of the "signature terms" at time t up to lag $-l$ and order p (finite length l memory and degree p approximation) is given by the tensor $m(\mathbf{z})_t = \tilde{\mathbf{z}}_t(1)\otimes\cdots\tilde{\mathbf{z}}_t(d)\otimes\cdots\otimes\tilde{\mathbf{z}}_{t-l}(1)\otimes\cdots\otimes\tilde{\mathbf{z}}_{t-l}(d) \in T^{d(l+1)}(\mathbb{R}^{p+1}) := \{\sum_{i_1,...,i_{d(l+1)}}^{p+1} a_{i_1,...,i_{d(l+1)}}\mathbf{e}_{i_1}\otimes\cdots\otimes\mathbf{e}_{i_{l(d+1)}} : a_{i_1,...,i_d(l+1)} \in \mathbb{R}\}$, where for $x \in \mathbb{R}$, $\tilde{x} := (1, x, x^2, ..., x^p)$.
2. (Signature from state equation) As in the continuous case (Cirone, 2024), it is possible to obtain the monomial terms of $m(\mathbf{z})_t$ from a state equation involving a tensor order-lowering operator (to fix the size of the output) composed with a tensor product of the Vandermonde vector input, and scaling by a tempering coefficient λ to allow contraction in the state vector, which in turn grants the echo state property (injectivity) and fading memory property (continuity) of the associated solution mapping ("filter"). For instance, for $d = 1$, we can use $\pi_l : a_{i_1,...,i_l}\mathbf{e}_{i_1}\otimes\cdots\otimes\mathbf{e}_{i_l} \mapsto a_{1,i_2,...,i_l}\mathbf{e}_{i_2}\otimes\cdots\otimes\mathbf{e}_{i_l}$ (distribute to get action over $T^{l+1}(\mathbb{R}^{p+1})$) in the relation

$$\mathbf{x}_t = \lambda\pi_l(\mathbf{x}_{t-1}) \otimes \tilde{\mathbf{z}}_t + \bar{\mathbf{z}}_t, \mathbf{x}_t \in T^{l+1}(\mathbb{R}^{p+1}) \tag{3}$$

with $\bar{\mathbf{z}}_t = \mathbf{e}_1^{\otimes l} + \mathbf{z}_t\mathbf{e}_1^{\otimes(l-1)}\otimes\mathbf{e}_2$ (Cuchiero, 2021, Prop II.2). The solution at time t can be shown to be $Am(\mathbf{z})_t$, with $A \in L(T^l(\mathbb{R}^{p+1}), T^l(\mathbb{R}^{p+1}))$ invertible.
3. (Compressibility) Consider a state system given by a continuous state map $\mathbf{x}_t = F(\mathbf{x}_{t-1}, \mathbf{z}_t)$ going from $\mathbb{R}^N \times D_d \to \mathbb{R}^N$, where $D_d \subset \mathbb{R}^d$ is compact. It is assumed to be sufficiently contractive in the state vector so that it induces a ESP and FMP filter U.

If Q is a centered spanning set of the state space $\mathbb{R}^N$, and f is a Johnson-Lindenstrauss mapping set to preserve the isometry of Q up to some ϵ error, then the alternative state map $f^* \circ f(F(\cdot, \cdot))$ can be shown to induce a filter U' that is uniformly close to U, up to $C||f||_{\ell^2(\mathbb{R}^N)\to\ell^2(\mathbb{R}^N)}\epsilon^{1/2}$, with constant C depending on Q.
4. (Law of the compressed state matrices) The relation in 3 can be lifted to

the isomorphic space $\mathbb{R}^{(p+1)^{(l+1)}}$, with the actions of $\pi_l, \otimes \mathbf{e}_1, ..., \otimes \mathbf{e}_{p+1}$ corresponding to some matrices in $\mathbb{R}^{(p+1)^{(l+1)} \times (p+1)^{(l+1)}}$. Then, it remains to apply the compressibility result in bullet point 3) with an optimally chosen spanning set Q, and then show that applying f to the system, where f is chosen to be in the Gaussian ensemble, still yields state matrices which are Gaussian.

3 Over structured signals

So far, the approximation results above have been surprisingly "data agnostic," in the sense that we have not used any regularity conditions on our input signals other than the general requirement of absolute continuity (in continuous-time) or simple boundedness (in discrete-time). We try to devise an approach that stays close to the discrete-time framework in (Cuchiero, 2021).

One possible approach is to consider signals who, at every time t and up to some lag $-w$, remain (approximately) sparse in some fixed basis. If $(\cdot)_{t-w}^t : \ell^\infty(\mathbb{Z}^{\leq 0}; \mathbb{R}) \to \mathbb{R}^w$ denotes windowed truncation, i.e. $\mathbf{z}_{t-w}^t = (\mathbf{z}_{t-1}, ..., \mathbf{z}_{t-w})^T$, we will

1. (Sparsity up to lag $-w$) Assume the existence of an orthnormal basis $\Phi \subset \mathbb{R}^w$ such that $\Phi \mathbf{z}_{t-w}^t$ is s-sparse, for all $t \in \mathbb{Z}^{\leq 0}$.
2. (A tradeoff) Instead of seeking the signature terms $m(\mathbf{z})_t, t \leq 0$, we will contend ourselves with low order terms of $m(\mathbf{y})_t$, where $\mathbf{y}_t = \Phi \mathbf{z}_{t-w}^t$. The motivation is that given the coefficients $\Phi \mathbf{z}_{t-w}^t$, we should be able to approximate a regular system with input $\mathbf{z}$ having memory up to lag $-(w-1)$ given by $U(\mathbf{z})_t = F((\mathbf{z}_t, ..., \mathbf{z}_{t-w+1}))$, by a system with input $(\mathbf{y}_t)_t = (\Phi \mathbf{z}_{t-w}^t)_t$ having 0 memory given by $V(\mathbf{y}_t) = f(\mathbf{y}_t)$. Such systems with no memory have Taylor expansion given by:

$$V(\mathbf{s})_t = \sum_{j=0}^{\infty} \frac{1}{j!} \sum_{i_1,...,i_j=1}^{d} D^{i_1,...,i_j} f(\mathbf{0}) \mathbf{s}_{i_1} \cdots \mathbf{s}_{i_j},$$

which only use signature terms $\mathbf{s}_t(i_1) \cdots \mathbf{s}_t(i_j)$ up to lag 0. This is to be shown by a density argument.

In other words, this tradeoff is worthwhile: doing the work to get the coefficients $\Phi \mathbf{z}_{t-w}^t$ should ensure less work to attain the signature terms needed to approximate the target input-output system. More precisely, we will try

to mimick the proof in (, Theorem B.II.3) to show that it suffices to use systems with diagonal state matrices to approximate $V(\mathbf{y})$.
3. (Analysis filter as a state system) However, we might be able to win big by showing that retrieving $\Phi\mathbf{z}_{t-w}^{t}$ can be done cheaply as well! Returning to $d = 1$, note that the system $C(\mathbf{z})_t = \Phi\mathbf{z}_{t-w}^{t}$ is given by $C(\mathbf{z})_t = \Phi A\Phi^T C(\mathbf{z})_{t-1} + \Phi(\mathbf{z}_t, 0, ..., 0)^T$. where A is the shift-right operator having a subdiagonal of 1s and being 0 otherwise.
4. (Compressibility) We might have to verify contraction of the state map above, and otherwise scale the system with λ as done in Cuchiero's approach. Our system now having ESP and FMP, we will be able to guarantee that for a suitable Johnson-Lindenstrauss mapping f, the system $f^* \circ f(C(\mathbf{z})_t) = f^*(f \circ \Phi A\Phi^T C(\mathbf{z})_t + f \circ \Phi(\mathbf{z}_t, 0, ..., 0)^T)$ will also be ESP and FMP and (perhaps with high probability) within $g(\epsilon, s)$-error of $C(\mathbf{z})_t$ uniformly over $\mathbb{X} \times \mathbb{Z}^{\leq 0}$, where $\mathbb{X} \subset \ell^\infty(\mathbb{Z}^{\leq 0}; \mathbb{R})$ is our space of signals of interest, and $g(\epsilon, s)$ is to be derived using the RIP property of the Gaussian ensemble (Candes, 2008).

With the looser hypothesis of approximate sparsity, perhaps modeled by additive Gaussian noise ω, we will have to take into account a distortion in the RIP constant, which I think will be on the same order as the $||\omega||_2$.
5. (Law of compressed state matrices) We will then study the law of entries of the matrices $f \circ \Phi A\Phi^T$ and $f \circ \Phi(1, 0, ..., 0)^T$. I expect these to be (close to or exactly) Gaussian, since the multivariate Gaussian is rotation-invariant.
5. The final step will be to clarify that the tradeoff in Step 2, where we swapped knowledge of signature terms of $(\mathbf{z}_t)_t$ for knowledge of low order terms of $(\Phi\mathbf{z}_{t-w}^{t})_t$ can be done instead as such: swapping knowledge of signature terms of $(\mathbf{z}_t)_t$ for knowledge of low order terms of $(f \circ \Phi\mathbf{z}_{t-w}^{t})_t$ at the cost of some approximation error, given that we can switch our target input output system to $V(f^* \circ (\cdot))$.

6. (Generalization to $d > 1$) We must generalize to the case $d > 1$ of d-dimensional controls. I expect sparsity of signals across the time t to be translated into sparsity of signals across both the time t and the index $1 \leq i \leq d$.

Some remarks:
Remark 1. If the law of the entries of the compressed state matrices can be shown to be (approximately) Gaussian, then in effect this will imply that knowledge of the basis Φ is not required!!!
Remark 2. If all is well with this outline, it may be worthwhile to generalize

this to the case of continuous-time using Cirone's framework.